

Pyrotechnic electronics is here!
Build, see, and hear our...

FLAME SPEAKER

Striking, stupendous, stunning—this
is one project you'll never forget!

By Jorma Hyypia



Line your speaker cabinet with asbestos before installing this hot-fi super-fidelity loudspeaker! Otherwise your audio system will be burned to a cinder because—believe it or not—the speaker is a high-temperature gas flame! We call it the *Flame Speaker*!

We don't seriously suggest that this revolutionary speaker should replace your present loudspeaker. But you will have to look long and hard to find a more spectacular audio experiment for a high school physics project, a science fair demonstration, or for simply astounding friends in your home.

The Flame Speaker is more than a far-out stunt. Its inventors already foresee many applications of the principles involved; they range from new communications systems to important new rocket design techniques, and from fire alarm systems to musical candles and fireplaces.

Credit for the original idea goes to researchers W. Babcock, A. G. Cattaneo and K. L. Baker who stumbled on the curious phenomenon while investigating methods of underwater sound generation at the United Technology Center at Sunnyvale, California. Using their original report as a guide, the author set up a similar system to find out how the Flame Speaker can be made

to work! The results were astonishing. Absurdly simple equipment can be used to demonstrate the principles involved; only slightly more sophisticated equipment is needed to produce sound of excellent quality.

The UTC researchers claim that high frequency response of the Flame Speaker is better than what can be obtained with the most expensive conventional loudspeakers now available. As a super-tweeter, the Flame Speaker probably has no peer.

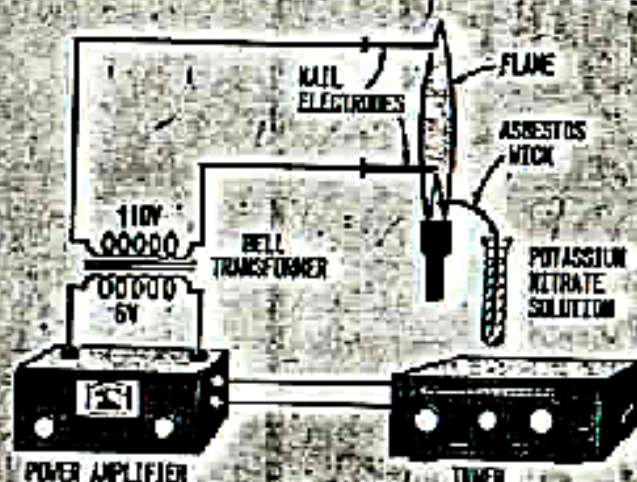
Bass response starts to drop off around 2000 to 3000 Hz at a rate of 6 dB per octave, though there is some degree of response all the way down to DC. But, this is not believed to be an inherent shortcoming of the system; low-frequency response is expected to improve markedly with the use of larger flames.

Now It Works. To make a musical flame, simply feed the output of a power amplifier (connected to a tuner, record player or tape recorder) to a step-up transformer and then to two electrodes positioned in the flame. A water solution of potassium nitrate must be fed into the flame to provide ionizable molecules for reasons that will become clear later.

This simple set-up will fill a large room with sound. Though there will be considerable distortion, the results will be so promi-

ing that you won't be able to resist cleaning up the sound by adding a DC bias circuit. When that is done, you will feel amply rewarded for the effort; the music emanating from the flame will be astonishingly sweet and clear.

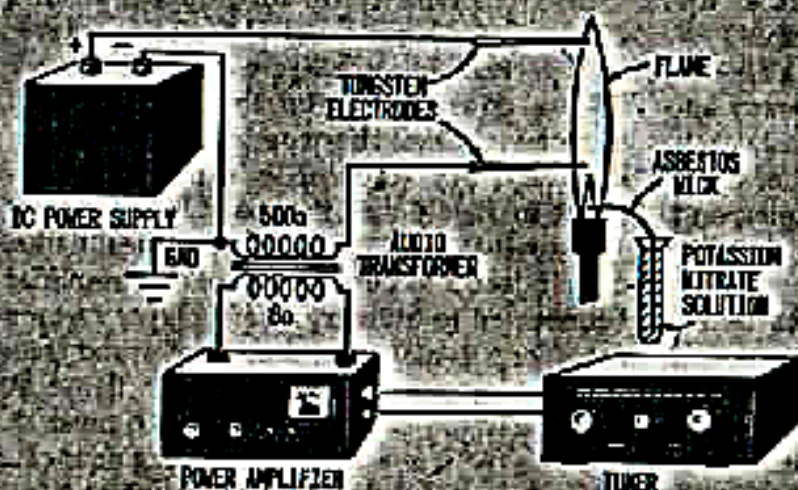
Though there is still much to be learned



Simple setup utilizing bell transformer and metal electrodes produces loud sound. Lack of DC bias voltage, however, leads to considerable distortion.

about the Flame Speaker, it seems clear that the sound reproducing abilities of a flame are related to the formation of *plasma*—a gas made up of positively charged atoms and free electrons. The gas used to create the flame does not have enough ionizable molecules to form a good plasma; hence the easily-ionized potassium nitrate is needed.

When an oscillating electric field is applied to the flame by means of the two electrodes,



Elaborate version above ensures excellent sound quality. Addition of audio transformer and DC bias voltage gives flame a real boost in its ability to reproduce audio. At right, lower electrode goes just above inner cone of flame; upper electrode is positioned for best sound.

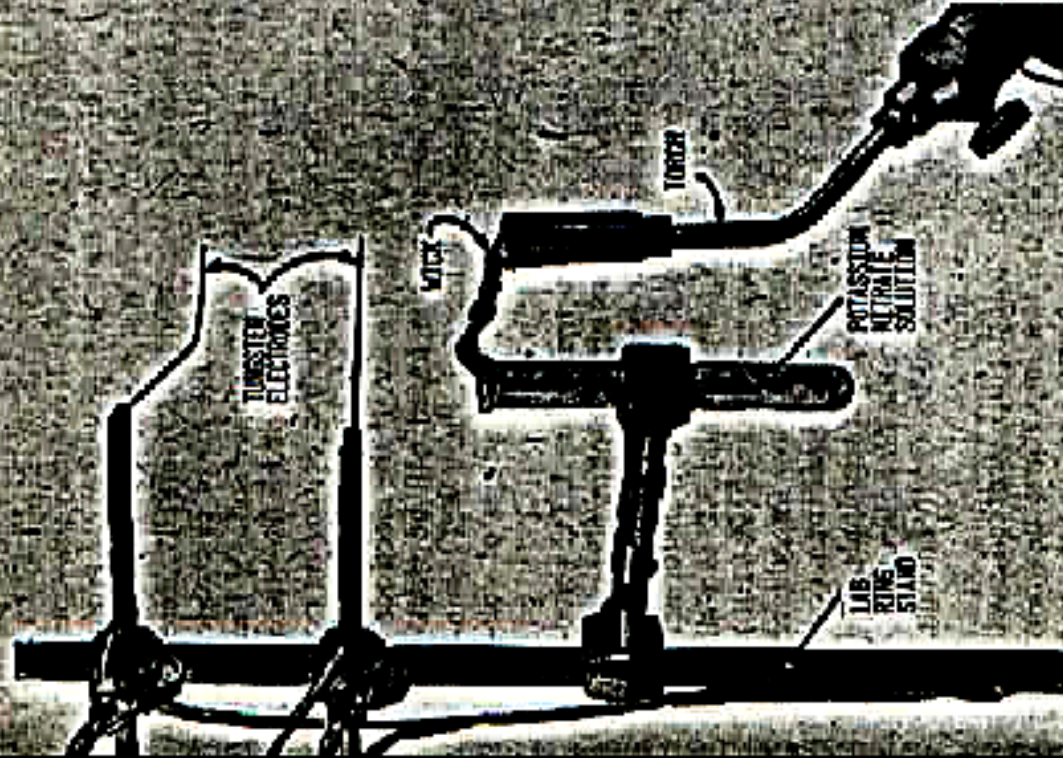
with the negative pole. Since positive ions are heavier than electrons, there is a net force in the direction of the negative pole. Thus the flame is bent by the electric field. This bending (vibration) of the flame is a function of the fluctuating electrical input. The flame remains electrically neutral because there are an equal number of plasma electrons and positive ions; the phenomenon is related only to the relative mobility of the electrons and positive ions.

We are so conditioned to think of sound generation in terms of vibrating *solids*—in particular the paper cone of a conventional loudspeaker—that it is at first difficult to imagine that a gaseous system might work in much the same way. For more than a century it has been known that flames can *respond* to sound waves, and can act as *amplifiers* of vibrations produced by other membranes. For the first time there is proof that a flame can also *generate* sound waves.

Give Me Some Skin. For some time scientists have speculated that flames, like liquids, exhibit a surface tension. We are all familiar with the surface tension "skin" on water that makes it possible to float a metal needle on water. The skin is not, of course, a physical entity that can be peeled off, like the skin of an onion. It only acts like a thin skin might act if it existed. The phenomenon results from molecular attractions that tend to pull the water into a shape consistent with the smallest possible volume.

The tapered shape of an undisturbed flame is probably due largely to this surface tension phenomenon. In fact, this flame speaker be-





Laboratory ring stand and clamps come in handy for holding tungsten electrodes and test tubes. Bend upper electrode only when narrow gap is desirable.

havior tends to confirm the flame "skin" notion.

We might now think of the flame "skin" as a sort of gaseous membrane that is in intimate contact with the surrounding air. As this skin vibrates, it moves the air in much the same way that a conventional speaker cone compresses and rarefies air to produce sound waves. However, the conventional speaker diaphragm is made of solid material that necessarily suffers from such factors as internal friction and mass inertia. No such diaphragm can ever be wholly efficient. On the other hand, a flame can be thought of as a virtually massless and frictionless membrane. It is able to vibrate much more freely than could any diaphragm made of solid material.

This explains why the flame is such an unusual "efficient tweeter." Why doesn't it woof at all? Probably only because all experiments to date have been performed with

haps far better than any conventional woofer.

How large would the flame have to be to be a good woofer? Only further experimentation will yield the answer. Meanwhile, we can't help but wonder what would happen if an intrepid audio bug were to stick electrodes into a burning oil well; he might succeed in wooing the Texans right off the map, or perhaps catalyze an earthquake.

Hi Heat, Hot Fl: It takes a lot of heat to jiggle electrons off atoms to create a really good plasma, so the UTC researchers used an oxy-acetylene welding torch in their experiments. Actually, even the 3000 degree temperature of such a torch produces at best a rather mediocre plasma, but one adequate for the purpose at hand.

In fact, you can get very good results even if a welding torch is not available. A bottled-gas shop torch (Bernz-O-Matic) serves well. Other flame sources worth trying include a plumber's blow torch, laboratory Bunsen burner, or an air-gas torch of the type used to blow glass in laboratories. As a matter of fact, the author managed to get a few low-fo squeaks using only the flame of a candle!

The potassium nitrate can be obtained from the local druggist or from a hobby shop selling chemicals for home experimentation. The solution strength is not critical; just dump a little into water and it will work.

The solution must be fed continually into the bottom portion of the flame so that the vaporized chemical streams upward past both electrodes; this is easy to see because the potassium colors the flame a faint lavender. Also, moving the wick end about on the burner helps because the sound volume will rise and fall abruptly as the vapor moves to and from the electrodes. Incidentally, move the wet wick only with a wooden stick or some other non-conductor; otherwise you



ould, Editor,
Mark Prod.

P
6
p d r

Q
7
p d r

PERIODIC TABLE OF THE ELEMENTS

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1a	2a	3a	4a	5a	6a	7a	0	0/1a
1 H 1.00797 +1 -1							2 He 4.0026 0	K
3 Li 6.939 2.1	4 Be 9.0122 2.2					9 F 18.9984 2.7	10 Ne 20.183 2.8	K-L
11 Na 22.98976 2.8-1	12 Mg 24.304 2.8-2					17 Cl 35.453 2.8-7	18 Ar 39.948 2.8-8	K-L-M
19 K 39.0983 2.8-1	20 Ca 40.078 2.8-2	21 Sc 44.9559 2.8-2	22 Ti 47.88 2.8-2	23 V 50.9415 2.8-2	24 Cr 51.9961 2.8-2	35 Br 79.904 2.8-7	36 Kr 83.80 2.8-8	L-M-N
37 Rb 85.4678 2.8-1	38 Sr 87.62 2.8-2	39 Y 88.9058 2.8-2	40 Zr 91.224 2.8-2	41 Nb 92.9063 2.8-2	42 Mo 95.94 2.8-2	53 I 126.90547 2.8-7	54 Xe 131.29 2.8-8	M-N-O
55 Cs 132.90545 2.8-1	56 Ba 137.327 2.8-2	57 La 138.90471 2.8-2	58 Ce 140.12 2.8-2	59 Pr 140.90765 2.8-2	60 Nd 144.242 2.8-2	85 At 210 2.8-7	86 Rn 222 2.8-8	N-O-P
87 Fr 223 2.8-1	88 Ra 226 2.8-2	89 Ac 227 2.8-2						O-P-Q

Atomic Number
Symbol
Atomic Weight
Oxidation States
KEY TO CHART
Electron Configuration

Transition Elements

Group 8

*Lanthanides	58 Ce 140.12 2.8-2	59 Pr 140.907 2.8-2	60 Nd 144.24 2.8-2	61 Pm 144.91 2.8-2	62 Sm 150.36 2.8-2	63 Eu 151.96 2.8-2	64 Gd 157.25 2.8-2	65 Tb 158.924 2.8-2	66 Dy 162.50 2.8-2	67 Ho 164.930 2.8-2	68 Er 167.26 2.8-2	69 Tm 168.934 2.8-2	70 Yb 173.04 2.8-2	71 Lu 174.97 2.8-2	-N-O-P	
**Actinides	90 Th 232 16-10-2	91 Pa 231 20-9-2	92 U 238 21-8-2	93 Np 237 22-9-2	94 Pu 239 24-1-2	95 Am 243 25-1-2	96 Cm 247 25-9-2	97 Bk 247 27-3-2	98 Cf 251 28-4-2	99 Es 252 29-8-2	100 Fm 257 30-8-2	101 Md 258 31-4-2	102 No 259 32-8-2	103 Lw 261 32-9-1	104 — 32-10-2	-O-P-Q

Numbers in parentheses are mass numbers of most stable isotope of that element.

ELECTRONIC CONFIGURATION OF THE ELEMENTS

References: P. H. Spedding and A. H. Deane, Editors, *The Rare Earths*, John Wiley and Sons, Inc., Publishers, New York, 1961. R. F. Gould, Editor, *Lanthanide-Actinide Chemistry*, Advances in Chemistry Series, No. 71, American Chemical Society, Washington, D.C., 1967. Paper No. 14, Mark Fred, *Electronic Structure of the Actinide Elements*.

By Laurence S. Fowles

Atomic No.	Element	K		L		M		N		O		P		Q	
		1	2	3	4	5	6	7	8	9	10	11	12	13	14
		s	p	s	p	s	p	d	f	s	p	d	f	s	p
1	H	1													
2	He	2													
3	Li	2	1												
4	Be	2	2												
5	B	2	2	1											
6	C	2	2	2											
7	N	2	2	2	1										
8	O	2	2	2	2										
9	F	2	2	2	2	1									
10	Ne	2	2	2	2	2									
11	Na	2								1					
12	Mg	2								2					
13	Al	2								2	1				
14	Si	2								2	2				
15	P	2								2	2	1			
16	S	2								2	2	2			
17	Cl	2								2	2	2	1		
18	Ar	2								2	2	2	2		
19	K	2								3					
20	Ca	2								4					
21	Sc	2								4	1				
22	Ti	2								4	2				
23	V	2								4	3				
24	Cr	2								4	4				
25	Mn	2								4	5				
26	Fe	2								4	6				
27	Co	2								4	7				
28	Ni	2								4	8				
29	Cu	2								4	9				
30	Zn	2								4	10				
31	Ga	2								4	10	1			
32	Ge	2								4	10	2			
33	As	2								4	10	3			
34	Se	2								4	10	4			
35	Br	2								4	10	5			
36	Kr	2								4	10	6			
37	Rb	2								5					
38	Sr	2								6					
39	Y	2								6	1				
40	Zr	2								6	2				
41	Nb	2								6	3				
42	Mo	2								6	4				
43	Tc	2								6	5				
44	Ru	2								6	6				
45	Rh	2								6	7				
46	Pd	2								6	8				
47	Ag	2								6	9				
48	Cd	2								6	10				
49	In	2								6	10	1			
50	Sn	2								6	10	2			
51	Sb	2								6	10	3			
52	Te	2								6	10	4			
53	I	2								6	10	5			
54	Xe	2								6	10	6			
55	Cs	2								7					
56	Ba	2								8					
57	La	2								8	1				
58	Ce	2								8	2				
59	Pr	2								8	3				
60	Nd	2								8	4				
61	Pm	2								8	5				
62	Sm	2								8	6				
63	Eu	2								8	7				
64	Gd	2								8	8				
65	Tb	2								8	9				
66	Dy	2								8	10				
67	Ho	2								8	10	1			
68	Er	2								8	10	2			
69	Tm	2								8	10	3			
70	Yb	2								8	10	4			
71	Lu	2								8	10	4	1		
72	Hf	2								8	10	4	2		
73	Ta	2								8	10	4	3		
74	W	2								8	10	4	4		
75	Re	2								8	10	4	5		
76	Os	2								8	10	4	6		
77	Ir	2								8	10	4	7		
78	Pt	2								8	10	4	8		
79	Au	2								8	10	4	9		
80	Hg	2								8	10	4	10		
81	Tl	2								8	10	4	10	1	
82	Pb	2								8	10	4	10	2	
83	Bi	2								8	10	4	10	3	
84	Po	2								8	10	4	10	4	
85	At	2								8	10	4	10	5	
86	Rn	2								8	10	4	10	6	
87	Fr	2								9					
88	Ra	2								10					
89	Ac	2								10	1				
90	Th	2								10	2				
91	Pa	2								10	3				
92	U	2								10	4				
93	Np	2								10	5				
94	Pu	2								10	6				
95	Am	2								10	7				
96	Cm	2								10	8				
97	Bk	2								10	9				
98	Cf	2								10	10				
99	Es	2								10	10	1			
100	Fm	2								10	10	2			
101	Md	2								10	10	3			
102	No	2								10	10	4			
103	Lr	2								10	10	4	1		
104		2								10	10	4	2		

* Note irregularity.

PHYSICAL CONSTANTS OF INORGANIC COMPOUNDS (Continued)

No.	Name	Synonyms and Formulae	Mol. wt.	Crystalline form, properties and index of refraction	Density or spec. gravity	Melting point, °C	Boiling point, °C	Solubility, in grams per 100 cc		
								Cold water	Hot water	Other solvents
229	Gallium									
230	chloride, tri-, hexammine	$\text{GaCl}_3 \cdot 6\text{NH}_3$	278.26					d	d	s NH ₃
231	chloride, tri-, monommine	$\text{GaCl}_3 \cdot \text{NH}_3$	183.11	wh. powd.	3.180 ²⁵	134		d	d	s NH ₃
232	ferrocyanide	$\text{Ga}[\text{Fe}(\text{CN})_6]_3$	914.74					d	d	l conc HCl
233	fluoride, tri-	GaF_3	126.72	wh. powd.	4.47 ± 0.01	subl 800 (in N ₂) - H ₂ O (vac) 140 - NH ₃ , 100	ca 1000	0.002		v a dil a; s (1)
234	fluoride, tri-, trihydrate	$\text{GaF}_3 \cdot 3\text{H}_2\text{O}$	180.78	wh. powd.					d	s dil H ₂ F ₄ v a dil HCl
235	fluoride, tri-, triammine	$\text{GaF}_3 \cdot 3\text{NH}_3$	177.81	wh. powd.					d	d
236	hydride	Digallane, gallmethane, Ga_2H_6	145.40	col. liq.		-21.4	139, d>130	d	d	d a, alk
237	hydroxide	$\text{Ga}(\text{OH})_3$	120.74	wh.		1440				
238	hydroxyquinoline deriv.	$\text{Ga}(\text{C}_6\text{H}_4\text{N})_3$	302.19	gray-yel. cr.		>150	subl vac	0.0001	0.0012	s dil a s a, alk; s (s)
239	iodide, tri-	GaI_3	450.43	dk. yel. cr.	4.15 ²¹	212 ± 1	subl 245		d	
240	iodide, tri-, hexammine	$\text{GaI}_3 \cdot 6\text{NH}_3$	552.02	wh. powd.					d	
241	iodide, tri-, monommine	$\text{GaI}_3 \cdot \text{NH}_3$	407.46	wh. powd.	3.035 ²⁴				d	
242	nitrate	$\text{Ga}(\text{NO}_3)_3 \cdot 3\text{H}_2\text{O}$		wh. cr. deq.		d 110	d to Ga_2O_3 200	v a	v a	s ahs al; i eth
243	nitride	Ga_3N_2	83.73	dk. gray powd.	6.1	subl 800				i dil a; s a h conc H ₂ SO ₄ , h conc NaOH
244	oxide, sesqui-(a)	Ga_2O_3	187.44	wh. hex. rhomb. 1.93, 1.90	6.44	1900, tr. to 8 500				s alk; v a s h a s a; s alk
245	oxide, sesqui-(b)	Ga_2O_3	187.44	monocl. rhomb.	5.88	1900(1740)				
246	oxide, sesqui-, monohydrate	$\text{Ga}_2\text{O}_3 \cdot \text{H}_2\text{O}$	208.45	wh. under cr. orthorhomb, 1.84	5.2	-H ₂ O, 400 tr to Ga_2O_3				
247	oxide, sub-	Ga_2O	165.44	blk. brn. powd.	4.77 ²¹	>600	subl >500			s a, alk
248	oxalate	$\text{Ga}(\text{C}_2\text{O}_4)_3 \cdot 4\text{H}_2\text{O}$	475.50	wh. micro cr., hyar oet.		-4H ₂ O, 180 d 200		0.1		
249	oxychloride	$6\text{Ga}_2\text{OCl}_3 \cdot 14\text{H}_2\text{O}$	179.25							v a KOH; i dil HNO ₃ ; s aet.
250	selenate	$\text{Ga}_2(\text{SeO}_4)_3 \cdot 16\text{H}_2\text{O}$	850.56	col. monoc. or tricl. cr.				57.54	v a	
251	selenide, mono-	Ga_2Se_3	148.08	dk. red-br. greasy leaf	5.03 ²⁴	980 ± 10				
252	selenide, sesqui-	Ga_2Se_3	370.32	red-b. brittle, hard	4.92 ²¹	>1020 ± 10				
253	selenide, sub-	Ga_2Se_3	318.40	bl.	5.02 ²⁴					
254	sulfate	$\text{Ga}_2(\text{SO}_4)_3$	427.62	wh. powd.		lime 16-19%		v a	v a	s al; i eth
255	sulfate, hydrate	$\text{Ga}_2(\text{SO}_4)_3 \cdot 18\text{H}_2\text{O}$	781.90	oet. cr.				v a	v a	s 60% al; i eth
256	sulfide, mono-	Ga_2S_3	101.78	yel. cr.	3.85 ²⁴	805 ± 10				s a, alk
257	sulfide, sesqui-	Ga_2S_3	230.03	yel. cr., or wh. amorph.	3.63 ²⁴	1255 ± 10				s a, alk
258	sulfide, sub-	Ga_2S_3	171.50	dk. gray	4.16 ²¹	d vac 800				s a, alk
259	telluride, mono-	Ga_2Te_3	197.32	blk. salt cr.	5.44 ²¹	824 ± 2				s a, alk
260	telluride, sesqui-	Ga_2Te_3	502.24	blk. brittle cr.	3.37 ²⁴	700 ± 2				
261	Germane									
262	bromo-	GeH_2Br_2	185.52	col. liq.	2.34 ²⁴	-52	52	d	d	s al; d alk
263	chloro-	GeH_2Cl_2	111.07	col. liq.	1.75 ²⁴	-52	28.0	d	d	s al; d alk
264	chloro trihydro-	GeH_3Cl	105.04	col. gas		-90.2	-20.0	d	d	s ahs al
265	dibromo-	GeH_2Br_2	234.47		2.80 ²⁴	-15.0	89.0	d	d	s al; d alk
266	dichloro-	GeH_2Cl_2	145.51	col. liq.		-68.0	19.5	d	d	s al; d alk
267	dichlorodifluoro-	GeCl_2F_2	181.48	col. gas	1.90 ²⁴	-51.8	-2.8	d	d	s ahs al
268	tribromo-	Germanium bromofluoride GeHBr_2	213.30	col. liq.		-24		d	d	d alk
269	trichloro-	Germanium chlorofluoride GeHCl_2	179.90	col. liq.	1.50 ²⁴	-71	78.2 d	d	d	d alk
270	trichlorofluoro-	GeCl_3F	197.43	col. liq.		-49.8	37.5	d		s ahs al
271	Germanium	Ge	72.59	gray-wh. met.-cutb.	5.30 ²⁴	937.1	2830			s h H ₂ SO ₄ , ac res; i alk

ICI
A: 11F
H: 11F
H: 11F

al a al
al a al
al a al

al eth
al a h com
al a h com
al a h a
al a h a

f: 1 d
a: 1 d
a: 1 d

h: 1 d

h: 1 d

g70	bromide, di-	GeBr ₂	232.41	sol need or pl		122	d	d	d	a. GeBr ₂ , al; 1 h
g76	bromide, tetra-	GeBr ₄	392.28	gray-wh. sol 1.6204	3.102 ²	20.1	196.5	d	d	a. sol, eth. liq; conc H ₂ SO ₄
g71	chloride, di-	GeCl ₂	143.90	wh. powd		d to Ge + GeCl ₄ -19.5	81	d	d	a. GeCl ₂ 1 al, eth
g72	chloride, tetra-	GeCl ₄	214.41	sol liq, 1.401	1.814 ¹⁰			d	d	a. sol, eth. v. a. d; HCl; conc HCl; conc H ₂ SO ₄
g73	fluoride, di-	GeF ₂	116.59	wh. cr, hygr.		d > 350				
g74	fluoride, tetra-	GeF ₄	148.58	sol. gas or liq. not liq at atm press	2.14 ¹⁰	subl - 87	subl		v. a.	
g75	fluoride, tetra-	GeF ₄ ·3H ₂ O	202.63	wh. cr, deliq.		d		d to GeO + H ₂ GeF ₆		
g76	hydride	Digermene, Ge ₂ H ₆	151.23	liq.	1.06 ¹⁰	-100	29, d 215	d		a. liq NH ₃
g77	hydride	Trigermene, Ge ₃ H ₈	228.83	sol liq.	2.2 ¹⁰	-105.6	110.5, d 190	d		a. CCl ₄
g78	hydride, tetra-	Germane, GeH ₄	76.62	sol gas	1.520 ¹⁰	-165	-88.5, d 350			a. liq NH ₃ , NaOH; al a. h. HCl
g79	nitride	Ge(NH) ₂	102.62	wh. amorph. powd		d 150			d to NH ₃ + GeO ₂	a. conc H ₂ , d; al; al a. CCl ₄ , eth; 1 CS ₂
g80	sulfide, di-	GeS ₂	326.40	or hex. pl.	5.37	d	subl var 240	a. d		a. conc H ₂ , d; al; al a. CCl ₄ , eth; 1 CS ₂
g81	sulfide, tetra-	GeS ₄	540.21	red-or, cub	4.322 ¹⁰	144	d 440	a. d		d al, conc; a. CS ₂ , CCl ₄ , liq, MeOH
g82	(tri-) nitride, di-	Ge ₃ N ₂	245.78	blk. cr.						
g83	(tri-) nitride, tetra-	Ge ₄ N ₃	273.80	wh. h. powd	5.25 ¹⁰	d 430				
g84	oxide, di-soluble	GeO ₂	104.00	trans	6.23 ¹⁰	1086 ± 3				a. alk
g85	oxide, di-soluble	GeO ₂	104.58	sol. hex. 1.550	4.328 ¹⁰	1115.0 ± 4		0.447 ¹⁰	1.07 ¹⁰	a. a. NaOH; 1 HCl; a. a. alk; 1 HCl; HF; one form NaOH, NH ₄ OH; one form Cl ₂ water, H ₂ O + NH ₄ OH; 1 a. alk all solv
g86	oxide, mono-	GeO	88.56	blk. cr. powd; 1.807		subl 710				a. a. NaOH; 1 HCl; a. a. alk; 1 HCl; HF; one form NaOH, NH ₄ OH; one form Cl ₂ water, H ₂ O + NH ₄ OH; 1 a. alk all solv
g87	oxychloride	GeOCl ₂	150.40	sol liq		-50				a. alk
g88	selenide	GeSe ₂	230.51	orange, rhomb(?)	4.58 ¹⁰	707 ± 3	d > 20	d	d	a. alk
g89	sulfide, di-	GeS ₂	186.72	wh. powd, or wh. orthorhomb	2.94 ¹⁰	ca 800	subl > 900	0.45 d	d to (FeO) + H ₂ S	a. alk; alk sulf; al, eth, aq 3.112 liq NH ₃
g90	sulfide, mono-	GeS	104.05	red-red amorph. or rhomb bipyram, blk	amorph: 3.01 rhomb: 4.01	530	subl 430	0.24		a. RCl, alk or alk sulf; al a. NH ₄ OH; 0.0473 liq NH ₃
g91	Germanium (Zintl)	See Beryllium Au	190.967	yel. duct. met., cub	19.3 liq; 17.0 ¹⁰	1053	2060			a. aq. reg. KCN, h H ₂ SO ₄ ; 1 a d al a. NaCN
g92	(I) bromide	AuBr	270.88	sol. blue-violet	7.8	d 115				a. aq. reg. KCN, h H ₂ SO ₄ ; 1 a d al a. NaCN
g93	(III) bromide	AuBr ₃	436.89	or cr. powd gray powd, or br. cr.		97.5-Hr, 190				a. eth. al
g94	(I) chloride	AuCl	232.42	yel. cr.	7.4	170 d to AuCl ₃	d 289.5	v. a. d	d	a. HCl, HBr
g95	(III) chloride	AuCl ₃ or AuCl ₄	303.33	orange-red or pr.	3.0	d 254	subl 265	68	v. a.	a. al, eth; al a. NH ₃ ; 1 CS ₂
g96	(I) cyanide	AuCN	222.98	li. yel. or powd	7.12 ¹⁰	d		v. a.	v. a.	a. KCN, NH ₄ OH; 1 eth, alk
g97	(III) cyanide	Cyanomeric acid, Au(CN) ₃ ·3H ₂ O or HAu(CN) ₄ ·3H ₂ O	329.07	sol. pl., hygr.		d 50		v. a.	d, v. a.	a. al, eth
g98	(I) iodide	AuI	323.87	greenish-yel. powd	5.25	d 120		v. a.	al a. d	a. KI
g99	(III) iodide	AuI ₃	577.08	dk. gro.	3.84	d 72		a. d		a. iodides a. HNO ₃
g100	(III) nitrate, hydrogen	Nitratous acid, AuH(NO ₂) ₂ ·3H ₂ O or H[Au(NO ₂) ₂]·3H ₂ O	500.04	sol. trist, oct						
g101	(III) oxide	Au ₂ O ₃	441.92			-0.100	-30, 250			a. HCl, conc HNO ₃ , NaCN
g102	(III) oxide	Au ₂ O ₃ ·H ₂ O				-11H ₂ O, 250		5.7 x 10 ⁻¹⁰ 26		a. HCl, NaCN, conc HNO ₃
g103	phosphide	Au ₃ P	485.80	gray	6.07					a. HCl, dil. HNO ₃
g104	selenide	Au ₂ Se	630.81		4.55 ¹⁰					a. aq. reg. KCN; 1 a al, eth, a. Na ₂ S
g105	(I) sulfide	Au ₂ S	629.00	br. blk. powd		d 340				
g106	(III) sulfide	Au ₂ S ₃	603.13	br. blk. powd	8.754	d 197		fresh sol	ppt. sol	

X-Ray Crystallographic Data of Minerals

Name and formula	Crystal system	Space group	Structure type	Z	a_0	b_0
Oxides and hydroxides—Continued						
157 Low-germanite GeO_2	tet.	$P4/mnm(130)$	rutile	2	4.3963 $\pm .0010$	
158 High-germanite GeO_2	hex.	$P3_121(152)$ $P3_221(154)$	α -quartz	3	4.087 $\pm .002$	
159 Ice H_2O	hex.	$P6_3/mmc(194)$		4	4.5212 $\pm .0010$	
160 Hafnia HfO_2	mon.	$P2_1/c(14)$	baddeleyite	4	5.1158 $\pm .0010$	5.1722 $\pm .0010$
161 Montroydite HgO	orth.	$Pnma(62)$		4	6.608 $\pm .003$	5.518 $\pm .003$
162 Periclase MgO	cubic	$Fm3m(225)$	rock salt	4	4.2117 $\pm .0005$	
163 Brucite $\text{Mg}(\text{OH})_2$	hex.	$P6_3ml(164)$	CdI_2	1	3.147 $\pm .004$	
164 Manganoite MnO	cubic	$Fm3m(225)$	rock salt	4	4.4448 $\pm .0005$	
165 Pyrolusite MnO_2	tet.	$P4/mnm(130)$	rutile	2	4.388 $\pm .003$	
166 Bixbyite Mn_2O_3	cubic	$Ia3(206)$	Ti_2O_3	16	9.411 $\pm .005$	
167 Hausmanite Mn_2O_3	tet.	$I4_1/amd(141)$		8	8.136 $\pm .005$	
168 Molybdate MoO_3	orth.	$Pbam(62)$		4	3.982 $\pm .002$	13.858 $\pm .005$
169 Bunsenite NiO	cubic	$Fm3m(225)$	rock salt	4	4.177 $\pm .002$	
170 Litharge PbO red	tet.	$P4/nmm(120)$		2	3.0760 $\pm .0040$	
171 Massicot PbO yellow	orth.	$Pb2a(32)$		4	5.480 $\pm .003$	4.765 $\pm .004$
172 Minium Pb_3O_4	tet.	$P4_2/mbc(135)$		4	8.815 $\pm .005$	
173 Senarmontite Sb_2O_3	cubic	$Fm3m(225)$	arsenic trioxide	16	11.152 $\pm .003$	
174 Valentinite Sb_2O_3	orth.	$Pern(56)$	antimony trioxide	4	4.914 $\pm .002$	12.408 $\pm .005$
175 Cervantite Sb_2O_4	cubic	$Fd3m(227)$		8	10.305 $\pm .005$	
176 Selenolite SeO_2	tet.	$P4_2/mbc(135)$ $P4_2/bc(100)$		8	8.35 $\pm .01$	
177 α -Quartz SiO_2^*	hex.	$P3_121(152)$ $P3_221(154)$		3	4.0136 $\pm .0001$	
178 β -Quartz SiO_2^*	hex.	$P6_322(181)$ $P6_622(180)$		3	4.999 $\pm .001$	
179 α -Cristobalite SiO_2	tet.	$P4_22_2(92)$ $P4_22_2(96)$		4	4.971 $\pm .003$	
180 β -Cristobalite SiO_2	cubic	$Fd3m(227)$		8	7.1982 $\pm .0010$	
181 Kestite SiO_2	tet.	$P4_22_2(92)$ $P4_22_2(96)$		12	7.450 $\pm .003$	
182 β -Tridymite SiO_2	hex.	$P6_32c(172)$ $P6_62/mmc(194)$		4	5.0403 $\pm .0020$	
183 Coesite SiO_2^*	mon.	$B2/b(15)$		16	7.152 $\pm .001$	12.370 $\pm .002$